

Smell and Weather Correlation Analysis

This notebook analyzes the relationship and correlation between smell reports and various weather variables: **temperature**, **pressure**, and **wind speed**.

```
In [1]: # Display the current Python environment details and search path
import sys
print(f"Python Path: {sys.executable}")
print(f"Searching in: {sys.path}")
```

```
Python Path: c:\Users\Nick\AppData\Local\Programs\Python\Python310\python.exe
Searching in: ['c:\\Users\\Nick\\AppData\\Local\\Programs\\Python\\Python310\\python310.zip', 'c:\\Users\\Nick\\AppData\\Local\\Programs\\Python\\Python310\\DLLs', 'c:\\Users\\Nick\\AppData\\Local\\Programs\\Python\\Python310\\lib', 'c:\\Users\\Nick\\AppData\\Local\\Programs\\Python\\Python310', '', 'c:\\Users\\Nick\\AppData\\Local\\Programs\\Python\\Python310\\lib\\site-packages']
```

```
In [2]: # Import essential libraries for data analysis and visualization
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import pearsonr

# Path to the cleaned weather and smell reports dataset
csv_path = "Weather and smell reports - Cleaned.csv"
```

1. Load and Clean the Dataset

We load the cleaned dataset using `pandas`. Column names are stripped of any leading/trailing spaces.

```
In [3]: # Load the cleaned dataset, strip trailing spaces from column names, and display
try:
    df = pd.read_csv(csv_path)
    # Clean column headers by stripping leading/trailing whitespace
    df.columns = df.columns.str.strip()
    print("Dataset preview:")
    display(df.head())
except FileNotFoundError:
    print(f"Error: Cleaned dataset file not found at: {csv_path}")
```

Dataset preview:

	smell_report_count	temperature	pressure	wind_speed
0	20	56	67	2
1	23	67	67	2
2	1	3	2	34
3	2	3	3	54
4	56	67	78	2

2. Statistical Correlation Matrix

Calculate the Pearson correlation coefficients between the number of smell reports and the various weather metrics.

```
In [5]: # Calculate Pearson correlation matrix focusing specifically on smell report
correlation_matrix = df[['smell_report_count', 'temperature', 'pressure', 'w

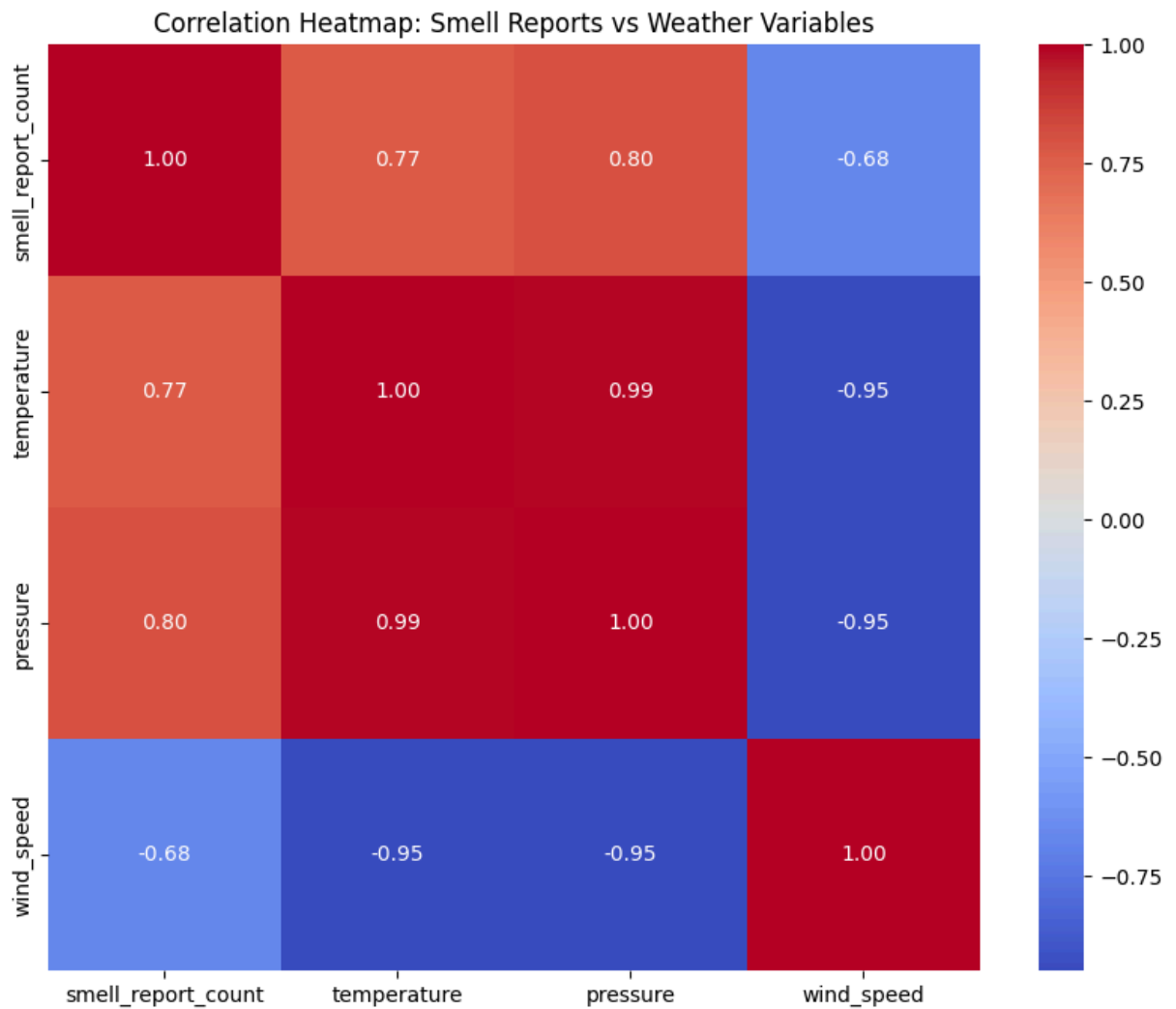
print("Correlation of weather variables with respect to smell reports:")
display(correlation_matrix['smell_report_count'].sort_values(ascending=False
```

```
Correlation Matrix:
smell_report_count    1.000000
pressure              0.801380
temperature           0.766748
wind_speed           -0.677272
Name: smell_report_count, dtype: float64
```

3. Visualization: Correlation Heatmap

Plotting a heatmap of the correlation matrix to visualize patterns.

```
In [6]: # Generate a correlation heatmap using Seaborn to visualize relationships
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap: Smell Reports vs Weather Variables')
plt.show()
```



4. Scatter Plots with Regression Lines

Generate regression plots for each weather variable to check for linearity and calculate p -values for statistical significance.

```
In [7]: # Define weather variables of interest
variables = ['temperature', 'pressure', 'wind_speed']
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Iterate through variables to generate scatter plots with linear regression
for i, var in enumerate(variables):
    # Plot regression line using Seaborn regplot
    sns.regplot(ax=axes[i], x=var, y='smell_report_count', data=df, scatter_kws={'s': 10})

    # Calculate Pearson correlation coefficient (r) and two-tailed p-value
    corr_val, p_val = pearsonr(df[var], df['smell_report_count'])
    axes[i].set_title(f'{var.capitalize()} vs Reports\n(r={corr_val:.2f}, p={p_val:.2f})')

plt.tight_layout()
plt.show()
```

